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Boiling pressure drop, local heat transfer distribution and critical heat flux in horizontal straight tubes

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ABSTRACT

The objective of the present work is to measure the heat transfer and pressure drop in flow boiling with refrigerant R123 as the working medium. Local heat transfer coefficient along the axial length of tube and overall averaged heat transfer coefficient are compared with available correlations. In the present study, spatial variation in heat transfer coefficient along the axial length of tube is analysed with the increase of quality. Variation in heat transfer coefficient in different flow boiling region *i.e.* subcooled, nucleate and convective is studied. The experiments are conducted with seven test sections made of SS304 having four different tube diameters 5.5 mm, 7.5 mm, 9.5 mm and 12.0 mm with two different heated lengths 500 mm and 1000 mm. The system pressure is varying from 1.3 bar to 3.2 bar. The experimental data of two-phase pressure drop and critical heat flux are compared with available correlations. Correlations are suggested to measure the local heat transfer coefficient during boiling process, two-phase pressure drop and critical heat flux. Data of critical heat flux are published to contribute in the data bank for horizontal straight tubes.

1. Introduction

The use of horizontal straight tubes in heat transfer devices is important in heat exchangers, evaporators, boilers and nuclear reactor. World's energy demand, fast depletion of fossil fuel resources and climate changes leads the investment in nuclear power and clean energy. Pressurise heavy water reactor (PHWR) has horizontal fuel channel alignment in nuclear reactor. Solar thermal energy is an attractive option as a renewable source of energy for power generation. Concentrating solar thermal power uses horizontal straight tube configuration for heat transfer. In the practical applications, flow boiling is the heat-transfer mode, where the highest heat transfer coefficient is achieved. Hence, it is essential to have well established correlation for boiling heat transfer coefficient. Most of the correlations available in the literature are developed for overall averaged heat transfer coefficient. To design a heat transfer device, it is important to measure different parameters at any local point. Rate of heat transfer at a given location is dependent on the spatial heat transfer coefficient at that location. In a flow boiling process, different heat transfer regimes are associated which changes with local quality. Heat transfer coefficient in different regimes of boiling like subcooled boiling, nucleate boiling and convective boiling are different. Hence, to improve the design of heat transfer devices, it is necessary to develop a correlation that predicts local heat transfer coefficient accurately. In the present work, local heat transfer coefficient is measured along the axial length of a straight tube with the increase in quality. Experimental local heat transfer coefficient is compared with the available correlations. A correlation is suggested to predict not only overall averaged heat transfer coefficient but also local heat transfer coefficient in different flow boiling regions.

Critical heat flux (CHF) sets the limit to exchange the heat in different industrial applications. With excessive heat supplied, boiling creates burnout of the tube either because of departure from nucleate boiling or dryout of liquid film. Hence, the ability to predict CHF during boiling process is of great practical importance. Numerous studies on CHF in straight vertical tubes are reported in literature. There are well developed look-up tables available to measure CHF and many methods to predict the CHF in vertical tubes. However, literature on CHF in horizontal tube is very less compared to that on vertical tubes. Number of CHF data on horizontal tube is very less compared to that of vertical tubes. Hence, there is no data bank or general correlation is available for horizontal tubes unlike vertical tubes. Literature survey is carried out on CHF in straight horizontal tubes. Details of each work along with the fluid used and corresponding operating and geometric

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Nomenclature

Symbol	Definition [Unit]
Cp	specific heat at constant pressure [J/kg K]
d	tube diameter [m]
G	mass flux [kg/m^2 s]
g	acceleration due to gravity [9.81 m^2/s]
h	heat transfer coefficient [W/m^2 K]
i	enthalpy [J/kg]
I	current [A]
k	thermal conductivity [W/m K]
L, l	length [m]
$\dot{m}$	mass flow rate [kg/s]
P	pressure [N/m^2]
p	periphery [m]
P_r	reduced Pressure (P_{sys}/P_{cr})
Q	heat supply [W]
q''	heat flux [W/m^2]
T	temperature [°C]
V	voltage [V]
x	quality of steam
Greek	
μ	dynamic viscosity [Ns/m^2]
ρ	density [kg/m^3]
σ	surface tension [N/m]
Subscript	
acc	acceleration
b	bulk
cr	critical
d	dryout
e	exit
fg	liquid to vapour
$fric$	friction
g	vapour
H	horizontal position
h	heated
i, in	inlet
l	liquid
o	outer
sat	saturated
sc	sub-cooled
sys	system
TP	two-phase
V	vertical position
w	wall
W	wetted
Abbreviation	
CHF	critical heat flux [W/m^2]
HTC	heat transfer coefficient [W/m^2 K]
Dimensionless number	
Bo	Boiling number, $Bo = q''/Gi_{fg}$
Ja	Jacob number, $Ja = Cp(T_{sat} - T_{in})/i_{fg}$
Pr	Prandtl number, $Pr = \mu Cp/k$
Re	Reynolds number, $Re = Gd/\mu$

parameters used to derive correlation of each work are presented in Table 1.

Comprehensive literature review on boiling heat transfer coefficient and two-phase pressure drop is presented in Hardik and Prabhu [14]. Hardik and Prabhu [14] collected the correlations available in literature on boiling heat transfer coefficient and two-phase pressure drop. Kandlikar [17] presented quantitative flow boiling curve which represents heat transfer coefficient with increase of quality. The heat transfer coefficient varies with increase of quality. The curve changes it slope in different regions of flow boiling *i.e.* subcooled flow boiling, nucleate flow boiling and convective flow boiling. The boiling curve describes the boiling process for different liquid to vapour density ratio fluids. Kandlikar [17] shows that the boiling process varies with variation in liquid to vapour density ratio. Almost all the available literature derived the correlations using different number of data points. Till now there is no work is reported to compare the correlation with complete flow boiling profile.

The experiments are conducted with seven test sections made of SS304 having four different tube diameters 5.5 mm, 7.5 mm, 9.5 mm and 12.0 mm with two different heated lengths 500 mm and 1000 mm. The wall thickness of all the test section is 0.25 mm. The system pressure is varying from 1.3 bar to 3.2 bar. Spatial variation in heat transfer coefficient along the axial length of tube is analysed with increase of quality. Variation in heat transfer coefficient in different flow boiling regions *i.e.* subcooled, nucleate and convective is studied. The correlations available in literature for boiling heat transfer coefficient, pressure drop and critical heat flux are used for refrigerant R123 fluid to check the effectiveness of the correlations. Correlations are suggested to predict the local heat transfer coefficient during boiling process, two-phase pressure drop and critical heat flux. Data of critical heat flux would serve in a data bank for horizontal straight tubes.

2. Description of the experimental rig

A schematic diagram of the experimental facility is shown in Fig. 1. It is a closed loop well insulated flow system with refrigerant R123 serving as a working fluid. The test facility consists of an insulated fluid reservoir, magnetically coupled sealless gear pump, pre-heater, condenser, ball valves and straight horizontal tube test section. Refrigerant R-123 has high boiling point temperature. Due to its high boiling point, water is used as secondary fluid to condense vapor in condenser and repeat the cycle. Secondary water circuit is open loop cycle. This makes system less complex, more stable and required less time to reach steady state. The gear pump with a mass flow rate range from 0 to 360 g/s is driven by a D. C. motor. The speed of the D. C. motor is varied between 0 and 3500 rpm by a motor controller. The temperature, pressure and flow rate measuring devices are instrumented with system to measure the pressure drop and heat transfer coefficient in the test sections. Coriolis mass flow meter (Make: Emerson, Sensor model: CMF025, Transmitter model: 1700R) is used to measure the mass flow rate of fluid. Coriolis mass flow meter has facilities to measure density of fluid and bulk fluid temperature. The inlet and outlet bulk fluid temperatures are measured with K-type thermocouples fixed on the surface of copper tubes. Infra-Red thermal camera (Make: Themoteknix, Model: VisIR640) is used to measure the wall temperature of the test section without disturbing test sections. The test section is painted with a thin coat of high temperature black board paint to have a uniform emissivity of 0.85. The differential pressure transmitters are used to measure the pressure drop across test sections and the pressure transmitters to measure the system pressure at the inlet and exit of the test section. Pressure transmitters and differential pressure transmitters are connected with the common pressure taps at inlet and exit of test section. Inlet tap is drilled 100 mm before the test section and exit tap is

Table 1
Available correlation for CHF in straight horizontal tubes.

Authors (Year)	*Geometric Parameters* d (mm), L (mm),	*Operating Parameters* q'' (kW/m^2); P (bar) G (kg/m^{2}sec)	Remark; Correlation
[22] Merilo (1979)	d = 5.6, 12.6, 19.1 L/d = 112 - 571	G = 700 - 8100 x_{in} = -0.35 - 0 ρ_l/ρ_g = 13 - 20.5	Uniform heat flux; Horizontal; Fluids: Water, R12
			$\frac{q''}{Gi_{fg}} = 575Re_l^{-0.34}\{Z^3B_o\}^{0.358}\left(\frac{\mu_l}{\mu_g}\right)^{-2.18}\left(\frac{l}{d}\right)^{-0.511}\left(\frac{\rho_l}{\rho_g}-1\right)^{1.27}(1-x_i)^{1.64}$ $Re_l = \frac{Gd}{\mu_l}$; $Z = \frac{\mu_l}{(\sigma d\rho_l)^{0.5}}$; $B_o = \frac{(\rho_l-\rho_g)gd^2}{\sigma}$; $x_i = \frac{Cp(T_i-T_{sat})}{i_{fg}}$
[34] Wojtan *et al.* (2005)	d = 13.6	G = 70 - 700 q'' = 2 - 57.5	Water heated refrigerants; Horizontal; Fluids: R-22, R410A
			$x_{d,e} = 0.61 \times Exp\left\{0.57 - 5.8\times10^{-3}\left(\frac{G^2d}{\rho_g\sigma}\right)^{0.38}\left(\frac{G^2}{\rho_g^2gd}\right)^{0.15}\left(\frac{\rho_g}{\rho_l}\right)^{-0.09}\left(\frac{q''_{CHF}}{q''_{Cr}}\right)^{0.27}\right\}$ $x_{d,i} = 0.58 \times Exp\left\{0.52 - 0.235\left(\frac{G^2d}{\rho_g\sigma}\right)^{0.17}\left(\frac{G^2}{\rho_g^2gd}\right)^{0.37}\left(\frac{\rho_g}{\rho_l}\right)^{0.25}\left(\frac{q''_{CHF}}{q''_{Cr}}\right)^{0.7}\right\}$ $q''_{Cr} = 0.131\rho_g^{0.5}i_{fg}\{g(\rho_l-\rho_g)\sigma\}^{0.25}$
[33] Wojtan *et al.* (2006)	d = 0.5 - 0.8 L = 20 - 70 L/d = 25 - 141	G = 400 - 1600 q'' = 3.2 - 600 ρ_l/ρ_g = 111.1 - 24.4	Uniform heat flux; Horizontal; Fluids: R134a, R245fa
			$\frac{q''}{Gi_{fg}} = 0.437\left(\frac{\rho_g}{\rho_l}\right)^{0.073}\left(\frac{G^2l}{\rho_l\sigma}\right)^{-0.24}\left(\frac{l}{d}\right)^{-0.72}$
[39] Zhang *et al.* (2006)	d = 0.33 - 6.22 L/d = 1 - 975	G = 5.33 - 134000 q'' = 94 - 276000 x_{in} = -2.35 - 0 P_{exit} = 1 - 190 bar	Database collection; Fluid: Water Horizontal, Vertical Shah's (1987) correlation works well for saturated flow boiling CHF Developed new correlation for saturated flow boiling CHF
			$\frac{q''}{Gi_{fg}} = 0.0352\left\{\left(\frac{G^2d}{\rho_l\sigma}\right) + 0.0119\left(\frac{l}{d}\right)^{2.31}\left(\frac{\rho_g}{\rho_l}\right)^{0.361}\right\}^{-0.295}\left(\frac{l}{d}\right)^{-0.311}\left\{2.05\left(\frac{\rho_g}{\rho_l}\right)^{0.17} - x_{in}\right\}$
Wu *et al.* (2011)	d = 0.223 - 2.98 L/d = 50 - 150	G = 23.4 - 4000 P_r = 0.005 - 0.25	Database collection; 629 data points from 16 different data sets; Seven fluids Fluids: Water, Nitrogen, R134a, R236fa, R245fa, R123, R12; Compared with Five correlations
[35]			$\frac{q''}{Gi_{fg}} = 0.6\left(\frac{l}{d}\right)^{-1.19}x_e^{0.817}$
Kim and Mudawar (2013) [18]	d = 0.51 - 6.0	G = 29 - 2303; P_r = 0.005 - 0.78 $Bo = 0.31\times10^{-4} - 44.3\times10^{-4}$ Re_L = 125 - 53770	Database collection; Horizontal, Vertical; 13 different fluids; 997 data points from 26 sources; Fluids: Water, CO2, R113, R134a, R22, R407C, R410A, FC72, R245fa, R290, R32, R1234yf, R1234ze,
			$x_{d,i} = 1.4\left(\frac{G^2d}{\rho_l\sigma}\right)^{0.03}\left(\frac{P_{sys}}{P_{Cr}}\right)^{0.08} - 15.0\left(\frac{q''}{Gi_{fg}}\frac{p_h}{p_W}\right)^{0.15}\left(\frac{\mu_lG}{\rho_l\sigma}\right)^{0.35}\left(\frac{\rho_g}{\rho_l}\right)^{0.06}$
[28] Shah (2015)	d = 0.13 - 24.3 L/D = 1.97 - 488	P_r = 0.005 - 0.9 G = 20 - 11390 x_{in} = -1.05 - 0.72	Database collection; Horizontal; 10 fluids; 878 data points from 39 data set from 18 sources Fluids: Water, FC-72, HFE-7100, R-12, R-32, R-113, R-134a, R-123, R-236fa, R-245fa; Zhange *et. al* (2006) works well, Merilo (1979) works well within its limitation
	$\frac{CHF_H}{CHF_V} = K_H$; $Fr_l = \frac{G^2}{\rho_l^2gd}$; $Fr_{TP} = \frac{x_{cr}G}{\{gd\rho_g(\rho_l-\rho_g)\}^{0.5}}$		$For\ x_{in} < 0,\ \ L/d < 10,\ \ K_H = 1$ $For\ x_{cr} \le 0.05,\ \ K_H = 0.725Fr_l^{0.082} \le 1$ $For\ x_{cr} > 0.05,\ \ K_H = 0.64Fr_{TP}^{0.15} \le 1$
[27] Shah (1987)	d = 0.315 - 37.5 L/d = 1.3 - 940	G = 4 - 29051 P_r = 0.0014 - 0.96 x_{in} = -4 - 0.85	Database collection; Vertical; 23 Fluids; 1443 data points from 62 Sources
	$Y \le 10^6, or\ L_E > \frac{160}{P_r^{1.14}}, use\ Bo_{UCC}$ $Y > 10^6, Min(Bo_{UCC}\ ;\ Bo_{LCC})$ $Y = \frac{G\cdot d\cdot Cp_l}{k_l}\left(\frac{G^2}{\rho_l^2gd}\right)^{0.4}\left(\frac{\mu_l}{\mu_g}\right)^{0.6}$ $Bo_{UCC} = 0.124\left(\frac{d}{L_E}\right)^{0.89}\left(\frac{10^4}{Y}\right)^n(1-x_{IE})$ $Bo_{LCC} = F_EF_xBo_0$	$Y \le 10^4, n = 0$ $Y \le 10^6, n = \left(\frac{d}{l_E}\right)^{0.54}$ $Y > 10^6, n = \frac{0.12}{(1-x_{IE})^{0.5}}$ $F_E = 1.54 - 0.032\frac{L_c}{d} \ge 1$	$x_i \le 0, L_E = L_c\ and\ x_{IE} = x_i$ $x_i > 0, L_E = L_B\ and\ x_{IE} = 0$ $\frac{L_B}{d} = \frac{x_{CHF}}{4Bo} = \frac{L_c}{d} + \frac{x_i}{4Bo}$ $F_3 = \left(\frac{1.25\times10^5}{Y}\right)^{0.833x_{CHF}}$
			$Bo_0 = Max\{15Y^{-0.612};\ 0.082Y^{-0.3}(1+1.45P_r^{4.03});\ 0.0024Y^{-0.105}(1+1.15P_r^{3.39})\}$
			$x_{CHF} > 0\ ;\ F_x = F_3\left\{1+\frac{(F_3^{-0.29}-1)(P_r-0.6)}{0.35}\right\}^C$; $P_r \le 0.6,\ C = 0$; $P_r > 0.6,\ C = 1$

drilled at 50 mm after the test section in disturbance free part of the experimental set-up. Thermocouples are attached on the other end of diagonal of pressure taps at inlet and outlet as shown in Fig. 1. During two-phase flow, temperature at the exit of the test section measured with thermocouple is equal to the saturation temperature corresponding to the pressure measured with the pressure transmitter. The range of the differential pressure transmitters is 0.622–62.2 kPa and 0–5 bar. The range of pressure transmitters are 0–10 bar.

All the test sections are heated electrically by passing DC current through the tube wall maintaining the uniform heat flux condition. DC power supply (Make: Aplab, Model: CVCC50 kW; 0–40 V

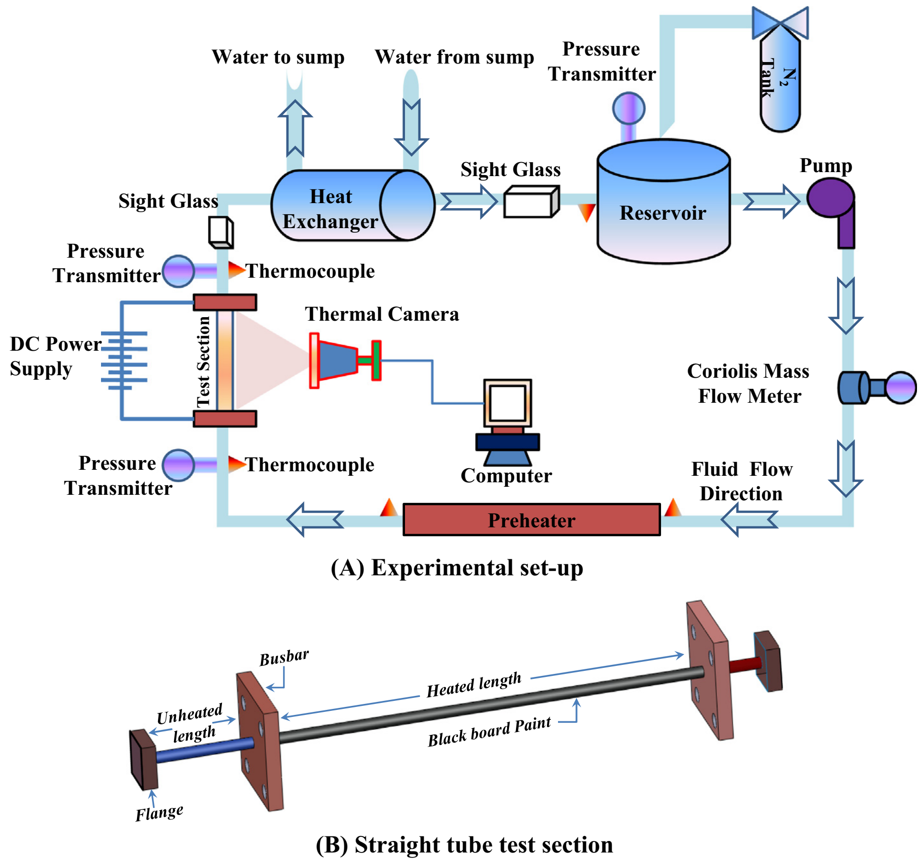

Fig. 1. Schematic of the experimental set-up.

voltage and 0–1250 A current) is used to supply power through the test section. The heat supplied is determined from the current and voltage across the test section. The current is measured using the power supply digital reading, while the voltage is measured using multimeter. Uniform inlet condition of the test section is maintained using pre-heater and condenser. Pre-heater is heated by passing AC current through Nichrome wire mounted on the wall of pre-heater. Heat supplied in pre-heater and test section is removed in condenser. Electromagnetic mass flow meter (Make: Manas, Model: SR1000P-DN25-PTFE) and K-type thermocouples are used to measure the heat absorbed by water. All the thermocouples are inserted into the fluid to measure the actual water temperature at inlet and outlet of the condenser. All the instruments are connected to Data Acquisition system (DAS) (Make: Atomberg, Model: AB4001) which is in turn connected to a computer. All the thermocouples are connected directly to DAS. The flow meter and pressure transmitters are connected to DAS through 100 Ω resistors to convert current signal (4–20 mA) into readable voltage signal (0.4–2 V). All the readings are taken after the complete system reaches steady state.

Straight tubes used for testing are made of stainless steel SS304 as shown in Fig. 1(B). The geometric details of the seven tubes tested and operating parameters covered in the present study are listed in Table 2. Four different tube diameters with two different heated lengths are used for analysis of effect of tube diameter and heated length. Number of critical heat flux data collected for specific test section is given in Table 2. The variables used in the present study are mass flux, heat flux, heated length, and tube diameter.

3. Data reduction

Hardik and Prabhu [15] suggested four definitions to predict CHF in test sections. In present study, non uniformity in the curve of heat flux versus wall temperature method is used for CHF prediction. During CHF experiments, heat flux is increased stepwise. After small increment in heat flux, the wall temperature from the thermal camera is captured. When the wall temperature of test section suddenly increases compared to previous wall temperature by at least 50 °C, it is determined as CHF. The experimental data obtained for CHF in all the test sections are presented in Appendix A. Following details are specified along with CHF data, Tube diameter, heated length of test section, system pressure which gives saturation temperature, mass flux, and exit quality.

3.1. Heat transfer coefficient

The heat supplied to the test section is calculated from the supplied electric current multiplied by the voltage across the test section as shown in Eq. (1). Heat loss from the test section for a given heat load is subtracted from the heat supply. Convective and radiation heat loss from the test section outer wall surface to the atmosphere are calculated theoretically. The procedure to calculate heat loss is given in Hardik and Prabhu [16]. Percentage of heat loss from test section outer surface depends on the wall temperature and total heat supply. In the present study, the percentage of heat loss is varying from 0.2% to 4% of total heat supply. The heat flux is calculated from Eqs. (2) and (3).

$$Q = VI \tag{1}$$

Table 2
Test sections geometrical and operating parameters.

Tube no	Tube characteristics			Operating characteristics					No. of CHF data points
	Inner diameter d (mm)	Outer diameter d_o (mm)	Heated length L_h (mm)	System pressure P_{sys} (bar)	Density ratio ρ_l/ρ_g	Mass flux G (kg/m^2 s)	Heat flux q'' (kW/m^2)	Quality x	
1	5.5	6	500	1.3–2.9	176–78	430–1500	55–263	−0.19 to 0.63	8
2	5.5	6	1000	1.3–3.2	176–70	400–1100	25–183	−0.21 to 0.82	7
3	7.5	8	500	1.3–2.9	176–78	440–1650	60–337	−0.19 to 0.56	10
4	7.5	8	1000	1.3–2.8	176–81	380–1200	30–180	−0.18 to 0.86	11
5	9.5	10	500	1.3–2.6	176–87	420–1300	50–328	−0.16 to 0.30	8
6	9.5	10	1000	1.3–2.7	176–84	400–1298	40–200	−0.16 to 0.39	5
7	12	12.5	1000	1.3–2.0	176–114	262–450	30–110	−0.10 to 0.55	6

$$Q_{conv} = Q - Q_{loss} \tag{2}$$

$$q'' = \frac{Q_{conv}}{\pi d L_h} = \frac{Q - Q_{loss}}{\pi d L_h} \tag{3}$$

The bulk fluid temperature at inlet and outlet of the test section is measured using thermocouples. The local bulk fluid temperature in subcooled region is measured by the linear interpolation between inlet temperature and saturation temperature. The saturation temperature at quality zero is based on inlet system pressure. The local bulk fluid temperature in saturated boiling region is calculated from inlet and outlet pressure. Interpolation profile is calculated from pressure drop profile. To draw pressure drop profile accurate, the value of constant 'C' is changed in the pressure drop correlation Eq. (14) such that experimental pressure drop is equal to the pressure drop from correlation. Local saturation temperature is calculated using the local pressure. Thermo-physical properties of liquid in the subcooled flow boiling condition are calculated at local bulk fluid temperature. In the saturated flow boiling region, the properties are calculated at the local saturation temperature. The vapour properties in all the calculations are calculated at local saturation temperature. All the thermo-physical properties used in data reduction are calculated from National Institute of Standards and Technology (NIST) tabulated values.

Local temperature distribution of the test section wall along the axial length is obtained by averaging ten thermal images for each configuration. Thermal images measure the outer wall temperature of the test section. Inner wall temperature is calculated from measured outer wall temperature using one dimensional, steady state heat conduction method with uniform volumetric heat generation across the test section wall due to electrical heating. The local quality and local heat transfer coefficient in subcooled and saturated boiling region are calculated from Eqs. (4)–(6). An Eq. (4) gives the quality at specific length of test section. The length of the test section from where, the subcooled flow boiling transits into saturated flow boiling (zero quality) is calculated from Eq. (7).

$$x_{local} = \frac{\frac{Q_{conv}}{\dot{m}}\frac{L_{local}}{L_h} - i_{in}}{i_{fg}} \tag{4}$$

$$i_{in} = C_p(T_{sat} - T_{in}) \tag{5}$$

$$h_{local} = \frac{q''}{T_{w,local} - T_{b,local}} \tag{6}$$

$$L_{(x=0)} = \frac{\dot{m} C_p (T_{sat} - T_{in}) L_h}{Q_{conv}} \tag{7}$$

Local heat transfer coefficient in a single phase turbulent flow is calculated from Dittus-Boelter correlation as given in Eq. (8). The local Reynolds number and local Prandtl number are calculated with properties taken at local bulk fluid temperature.

$$h_{l-local} = 0.023 Re_{local}^{0.8} Pr_{local}^{0.4} \frac{k_l}{d} \tag{8}$$

3.2. Pressure drop

The experimental pressure drop measured using differential pressure transmitter is the sum of single phase liquid pressure drop, subcooled boiling pressure drop, saturated boiling (diabatic two-phase) pressure drop and adiabatic two-phase pressure drop. Fig. 2 shows schematic diagram of the total pressure drop summed with pressure drop in different flow regions.

$$\Delta P_{Total} = \Delta P_l + \Delta P_{sc} + \Delta P_{TP} \tag{9}$$

Pressure drop in single phase flow is calculated using Blasius correlation. The subcooled flow boiling pressure drop is calculated using Baburajan et al. [1] correlation as given in Eq. (10). The tube diameter (d) is in mm and ΔP_{1P} is adiabatic single phase pressure drop with properties taken at saturated temperature. Single phase pressure drop and subcooled boiling pressure drop are generally much lower than two-phase pressure drop.

$$\frac{\Delta P_{sc}}{\Delta P_{1P}} = 1 + 32500 Bo^{1.6} Ja^{-1.2} \left(\frac{d}{9.53}\right) \tag{10}$$

The two-phase pressure drop in a flow boiling system is the sum of pressure drop due to fluid acceleration and frictional pressure drop as in Eq. (11). The gravitational pressure drop is not considered for horizontal tubes. The acceleration pressure drop is calculated from separated flow model and void fraction is calculated from Steiner equation (Didi et al. [9]) as shown in Eqs. (12) and (13);

$$\Delta P_{TP} = \Delta P_{fric} + \Delta P_{acc} \tag{11}$$

$$\Delta P_{acc} = G^2 \left\{ \left[\frac{(1-x)^2}{\rho_l (1-\alpha)} + \frac{x^2}{\rho_g \alpha} \right]_{out} - \left[\frac{(1-x)^2}{\rho_l (1-\alpha)} + \frac{x^2}{\rho_g \alpha} \right]_{in} \right\} \tag{12}$$

$$\alpha = \frac{x}{\rho_g} \left[(1 + 0.12(1-x)) \left(\frac{x}{\rho_g} + \frac{1-x}{\rho_l} \right) + \frac{1.18(1-x)\left[g\sigma(\rho_l - \rho_g)\right]^{0.25}}{G^2 \rho_l^{0.5}} \right]^{-1} \tag{13}$$

3.3. Uncertainties in the measured and calculated parameters

The uncertainties in the test section diameter and length are obtained by measuring the values at different locations. The uncertainties in mass flow rate, temperature, and pressure are calculated by comparing the reading of DAS with the value obtained from standard instruments. The uncertainties in heat flux resulted from the accuracy of the ampere meter and volt meter. The uncertainty in thermo-physical properties of liquid and vapour are calculated

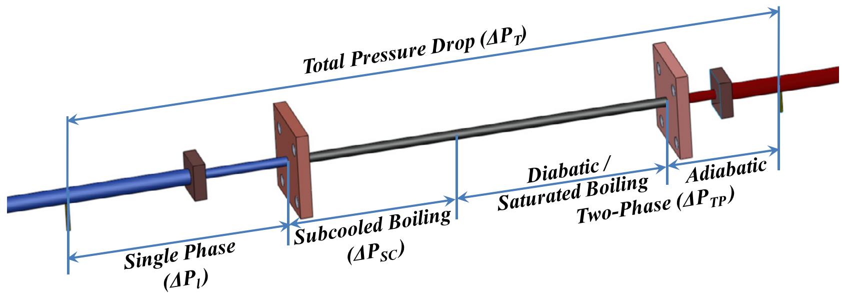

Fig. 2. Different regions of pressure drop.

Table 3
Experimental uncertainties.

Parameter	Relative uncertainty (%)	Parameter	Relative uncertainty (%)
Pipe diameter	1	Wall temperature	1.3
Pipe length	0.5	Mass flow rate	1.8
Current	0.5	Pressure	1.5
Voltage	1	Prandtl number	3.3
Bulk temperature	1.5	Reynolds number	3.6
Liquid density	1	Liquid friction factor	5.6
Liquid viscosity	3	Liquid Nusselt number	7.8
Thermal conductivity	1	Quality	3.4
Specific heat	1	Heat flux	3
Vapour density	1	Mass flux	2.7
Vapour viscosity	2	Boiling number	4
Surface tension	0.5	Heat transfer coefficient	11.4
Latent heat	0.5	Two-phase pressure drop	9.9

by comparing the correlations with the tabulated values of NIST. The uncertainty of different parameters used in the experimental analysis is shown in Table 3.

4. Results and discussion

Infra-red thermal imaging technique for measurement of wall surface temperature using thermal camera is validated with single phase flow of water in straight tube. The experimental local Nusselt number is calculated from local wall temperature and local bulk fluid temperature and compared with Dittus-Boelter correlation. The validation is given in Hardik et al. [13]. In the present section, the experimental set-up with fluid R123 is validated for single phase pressure drop and heat transfer coefficient. Wall temperature distribution along the axial direction of tubes is measured. Distribution of local heat transfer coefficient in subcooled and saturated (nucleate and convective) flow boiling regions is presented. Local experimental heat transfer coefficient is compared with available correlations along the axial length of the tube. Experimental data of overall averaged heat transfer coefficient, pressure drop and critical heat flux is compared with available correlations.

4.1. Validation of experimental set-up

Series of experiments are conducted to study the pressure drop and heat transfer coefficient in single phase flow. Measured experimental pressure drop and heat transfer coefficient are compared with standard well referred correlation to validate the experimental set-up. The pressure drop for all the test section shown in Table 2 is measured for Reynolds number ranging from 3000 to 30,000. The experimental friction factor is compared with Blasius correlation for turbulent flow. The mean absolute deviation between experimental data and correlation is less than 10%. The local heat transfer coefficient in single phase flow is calculated for all the test sections shown in Table 2. The measured heat transfer coefficient is compared with Dittus-Boelter correlation along the axial length of tube. Sample comparison is shown in Fig. 3. Fig. 3(A) shows the variation of wall temperature and liquid bulk fluid temperature. The experimental local heat transfer coefficient, calculated from measured wall temperature and bulk fluid temperature, is shown in Fig. 3(B). The flow becomes fully developed near the length to diameter ratio of 10. The mean absolute deviation between experimental data and correlation is less than 15% in fully developed flow region.

4.2. Flow boiling curve

Kandlikar [17] presented flow boiling curve. He suggested that the boiling curve presented by Collier [8] is qualitative curve and the slope of actual boiling curve is different than the one presented by Collier [8]. Different regions of boiling process depend on liquid to vapor density ratio. In this section, the flow boiling curve is developed from measured experimental local wall temperature. Wall temperature, bulk fluid temperature and local heat transfer coefficient along the axial length of tube are presented in Fig. 4 with increase of quality. Fig. 4(A) shows the distribution of wall and bulk fluid temperature with the quality and with non-dimensional axial length. The wall and bulk fluid temperatures are calculated with procedure as given in data reduction section. Heat transfer coefficient is calculated from bulk temperature and wall temperature. Heat transfer distribution is shown in Fig. 4(B) in the form of flow boiling curve. Fig. 4 shows the different regions of flow. Single phase heat transfer coefficient calculated from Dittus-Boelter correlation is shown in order to compare the enhancement in two-phase flow. The flow becomes fully developed within a non-dimensional length of 10. Boiling starts from the starting of test section due to high heat flux.

In the subcooled boiling region, the wall temperature is almost uniform and bulk fluid temperature increases. Hence, heat transfer coefficient increases. In the nucleate boiling region, the wall tem-

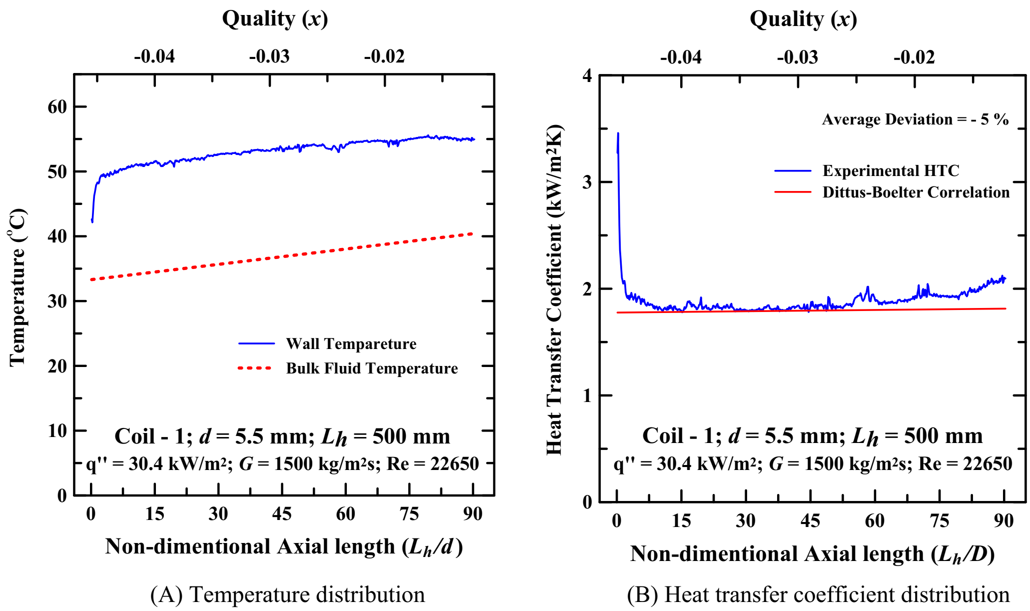

Fig. 3. Temperature and heat transfer coefficient distribution in single phase flow along the axial length of tube.

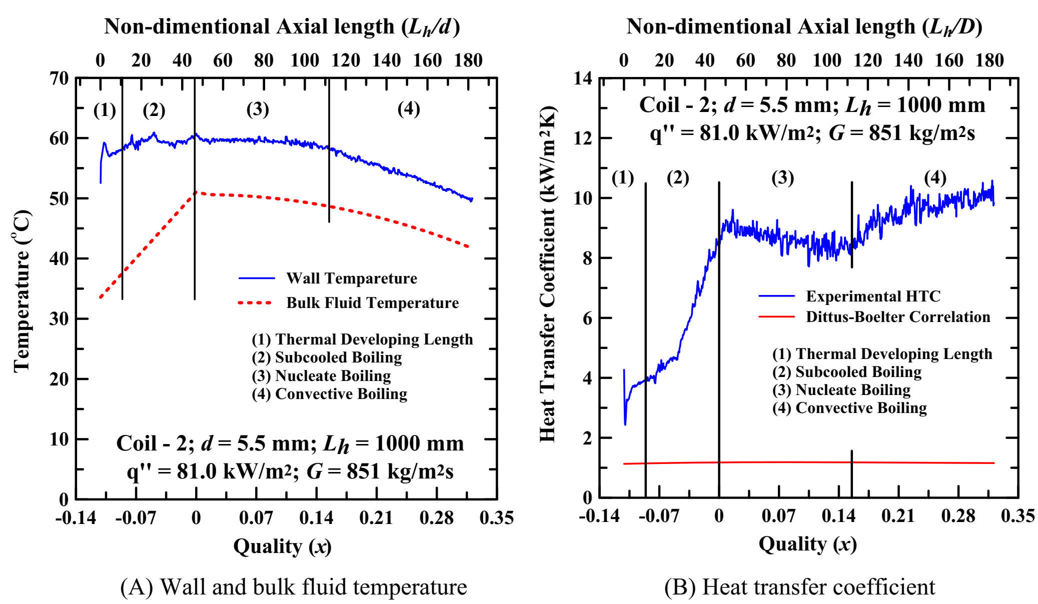

Fig. 4. Temperature and heat transfer coefficient distribution in different regions of flow boiling along the axial length of tube.

perature remains uniform while the bulk fluid temperature decreases due to pressure drop. The heat transfer coefficient decreases in the nucleate boiling region. The length of the nucleate boiling region is large. Hardik and Prabhu [14] showed that the length of nucleate boiling region is less than 0.08 quality for low pressure water system. In present case, the length of nucleate boiling region is nearly 0.16 quality. This is due to high density ratio of the present fluid compared to water and steam in Hardik and Prabhu [14] as suggested by Kandlikar [17]. The wall temperature and bulk fluid temperature decreases in convective boiling region. The heat transfer coefficient increases in convective boiling region. The enhancement of heat transfer coefficient in convective boiling region is less compared to the enhancement with water shown in Hardik and Prabhu [14]. This may be attributed to high density ratio.

4.3. Comparison of overall averaged heat transfer coefficient with the available correlations

The local heat transfer coefficient is measured along the axial length of tube. Overall averaged heat transfer coefficient is calculated by averaging the local heat transfer coefficient for whole length of tube. One data point represents the overall averaged heat transfer coefficient for one tube diameter at one specific mass flux and one specific heat flux. Total 171 cases of local heat transfer coefficient are analysed and based on that 171 data points for overall averaged heat transfer coefficient are collected. These data points are compared with general correlations available in the literature. The comparison of experimental data with some general correlations is shown in Fig. 5. The correlations given by Liu and Winterton [21] and Schrock and Grossman

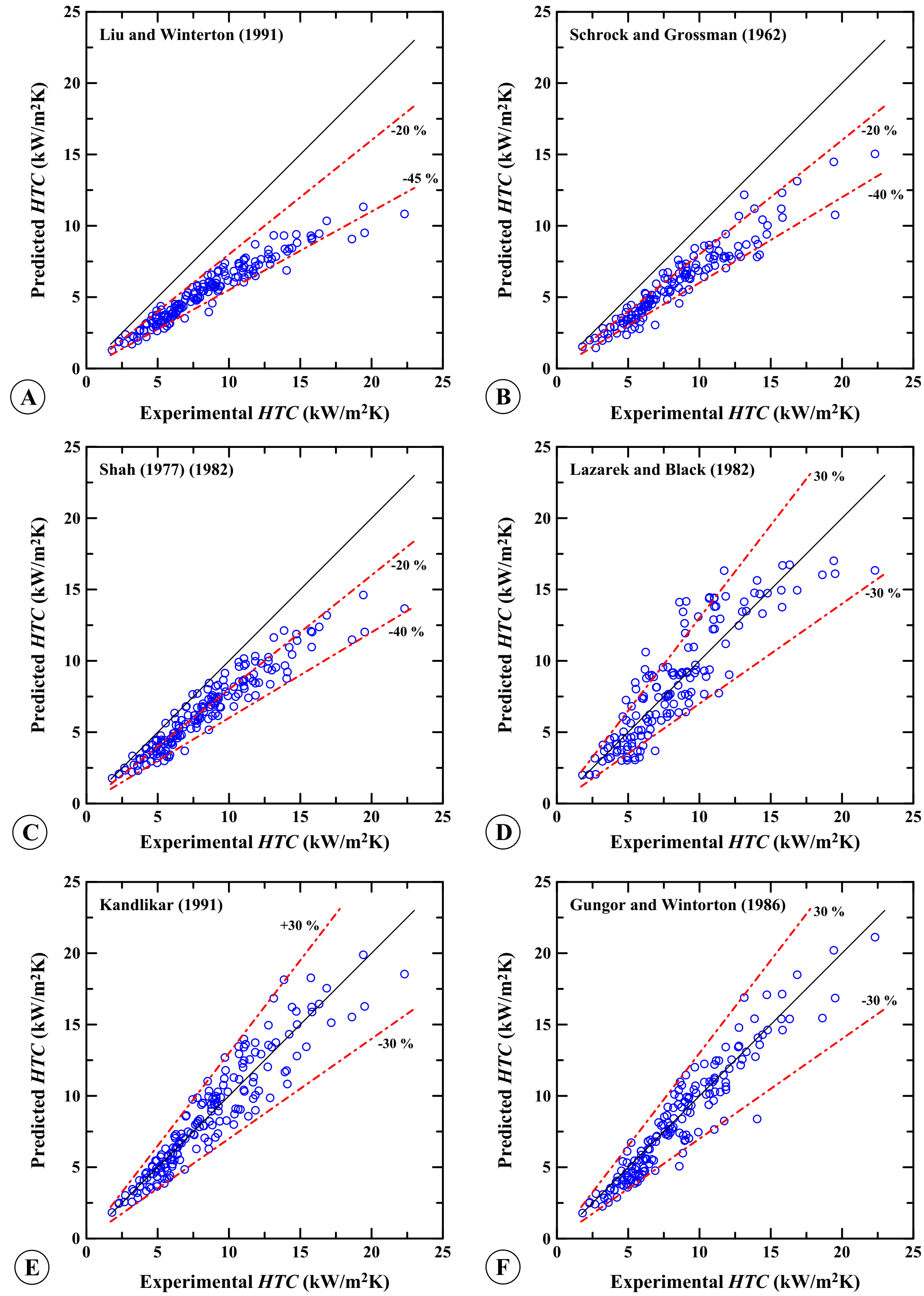

Fig. 5. Comparison of overall averaged experimental heat transfer coefficient with the available correlations.

[26] underpredicts the heat transfer coefficient for the present fluid. Most of the data points deviate from 0 to −40% with Shah [29,30] correlations. Experimental results are well centred with Lazarek and Black [20] correlation. Correlation measured 78% of data points within 30% deviation. The correlations like Kandlikar [17], Gungor and Winterton [12] match well with the experimental results. The mean absolute deviation with Kandlikar [17] correlation is 11.3% and with Gungor and Winterton correlation is 11.9%.

The statistical comparison between measured data and data predicted using correlations is given in Table 4. Comprehensive details of all the correlations analysed in Table 4 are available in Hardik and Prabhu [14]. Hardik and Prabhu [14] presented statistical analysis of all the correlation for flow boiling with water. Kandlikar [17] correlation and Shah [30] correlation predicted experimental results of water flow boiling. Shah correlation is not able to predict present experimental results accurately. Gungor and Winterton [12] correlation has large deviation with water flow

Table 4
Comparison of experimental results with correlations reported in the literature .

Sr. No.	Author	Absolute mean	Average	Maximum range
1	Kandlikar [17] (Saturated)	11.3	−0.96	−30 to 31
2	Gungor and Winterton [12] (Saturated)	11.9	−4.2	−41 to 29
3	Lazarek and Black [20]	19.0	0.84	−47 to 70
4	Shah [30]	21.7	−21.6	−47 to 4
5	Schrock and Grossman [26]	29.0	−29.0	−56 to −7
6	Chen [4]	30.3	−30.3	−52 to −12
7	Liu and Winterton [21] (Saturated)	35.1	−35.1	−54 to −13
8	Klimenko [19]	36.6	18.8	−31 to 54
9	Kandlikar [17] (Subcooled)	10.1	−4.7	−25 to 13
10	Gungor and Winterton [12] (Subcooled)	16.8	16.8	2 to 29
11	Liu and Winterton [21] (Subcooled)	17.0	−33	−49 to −13
12	Shah [29] (Subcooled)	22.2	−21.9	−45 to 4
13	Moles and Shaw [24] (Subcooled) ($x < -0.03$)	46.5	46.5	20 to 85

Absolute mean deviation = $1/N\sum_{i=1}^{N}[(|X_{cal} - X_{exp}|)/X_{exp}]$.
Average deviation = $1/N\sum_{i=1}^{N}[(X_{cal} - X_{exp})/X_{exp}]$.
Max range of deviation = $Min_{i=1}^{N}[(X_{cal} - X_{exp})/X_{exp}]$ to $Max_{i=1}^{N}[(X_{cal} - X_{exp})/X_{exp}]$.

boiling. But, Gungor and Winterton [12] correlation works well for present study. Kandlikar [17] correlation is working well for water as well as R123. Hence, it is better to use Kandlikar [17] correlation for flow boiling heat transfer coefficient measurement.

4.4. Comparison of local heat transfer coefficient with available correlations in mixed (Subcooled + Saturated) flow boiling

Flow boiling process includes subcooled boiling, nucleate boiling and convective boiling. The heat transfer coefficient in different boiling regions is different. The slope of heat transfer curve is changed from one region to another boiling region. The correlation should have to represent the boiling process in different regions. Most of the available literature shows the average heat transfer coefficient. The correlation that works well to predict overall averaged heat transfer coefficient need not essentially represent the boiling process precisely. Local heat transfer coefficient measured along axial length of tube is compared with the available correlations as a function of quality. A typical comparison for one sample case is shown in Fig. 6. The experimental heat transfer coefficient is measured in both subcooled and saturated regions. The correlations available for subcooled and saturated regions are compared with the experimental results in respective regions.

The correlations given by Liu and Winterton [21], Gungor and Winterton [12] and many others in from of single equation are showing continuous increment with increase of quality. Lazarek and Black [20] suggested the correlation to predict heat transfer coefficient in saturated region. The correlation is extended in subcooled region in present work. The correlation given by Lazarek and Black [20] remains uniform with increase of quality. The correlation given by Lazarek and Black [20] is function of Boiling number and Reynolds number. Hence, the correlation varies with variation in mass flux and heat flux and not with variation in quality. Lazarek and Black [20] correlation shows no change in heat transfer coefficient with increase of quality. Shah [30] and Kandlikar [17] gave correlations with two equations. Correlations represent different slopes for heat transfer coefficient in nucleate and convective boiling. Shah [30] shows a small length for nucleate boiling region. Overall, Kandlikar [17] correlation represents the boiling process.

Separate correlations for different boiling regions should have to intersect at transition region. The individual correlations developed for subcooled and saturated heat transfer coefficient should have to intersect at zero quality. Gungor and Winterton [12] gave separate correlations for subcooled and saturated boiling regions. These correlations have large deviation at transition region (At zero quality).

4.5. Comparison of critical heat flux with available correlations

Experimental CHF data are compared with most recently developed Shah [28] general correlation. The correlation is in form of correction factor to calculate CHF in horizontal tubes from vertical tube. CHF in vertical tube is calculated from Shah [27] correlation as suggested by Shah [28]. Shah [27] gave two correlations to predict CHF. One is upstream condition correlation (UCC) and second is local condition correlation (LCC). Shah [27] gave limitations to use either one of those correlations. Fig. 7(A) shows the comparison between present experimental CHF data and Shah [28] correlation. The correlation calculates the values with either UCC or LCC based on satisfied limitation condition. The correlation works well when the values predicted using UCC but it underpredicts the value when predicted using LCC. Hence, experimental data are compared with Shah [28] correlation with all the data predicted using UCC. The comparison is shown in Fig. 7(B). The correlation predicted data with reasonable accuracy compared to when predicted using LCC. The value of correction factor calculated using Shah [28] correlation varies from 0.8 to 1 for present study. Hence, the comparison between present horizontal tube CHF data and Shah [27] correlation will not deviate much. Experimental data are compared with Shah [27] vertical CHF correlation. Comparison is given in Fig. 7(C). Shah [27] correlation is not deviating much from experimental data.

The experimental CHF data are compared with other available CHF correlations. The comparison is shown in Fig. 8. Wojtan et al. [34] gave the condition for dryout completion quality. CHF is calculated from the dryout quality. The correlation underpredicts the experimental results of present study. Wojtan et al. [33] developed the correlation for micro channel. The correlation underpredicts the results. Hence, the correlation developed for microchannel might not be used for conventional tubes. Zhang et al. [39] developed correlation for minichannels with data available in literature. The correlation slightly overpredicts the results. Wu et al. [35] developed correlation for boiling number at CHF is function of only length to diameter ratio and exit quality. The correlation has two unknown parameters heat flux and exit quality. Hence, iterative method with initial assumption for exit quality 0.3 is used to solve equation. The correlation is not working well for present experimental data. Recent correlation developed by Kim and Mudawar [18] predicts the critical quality. The correlation has two unknown parameters heat flux and critical quality. The

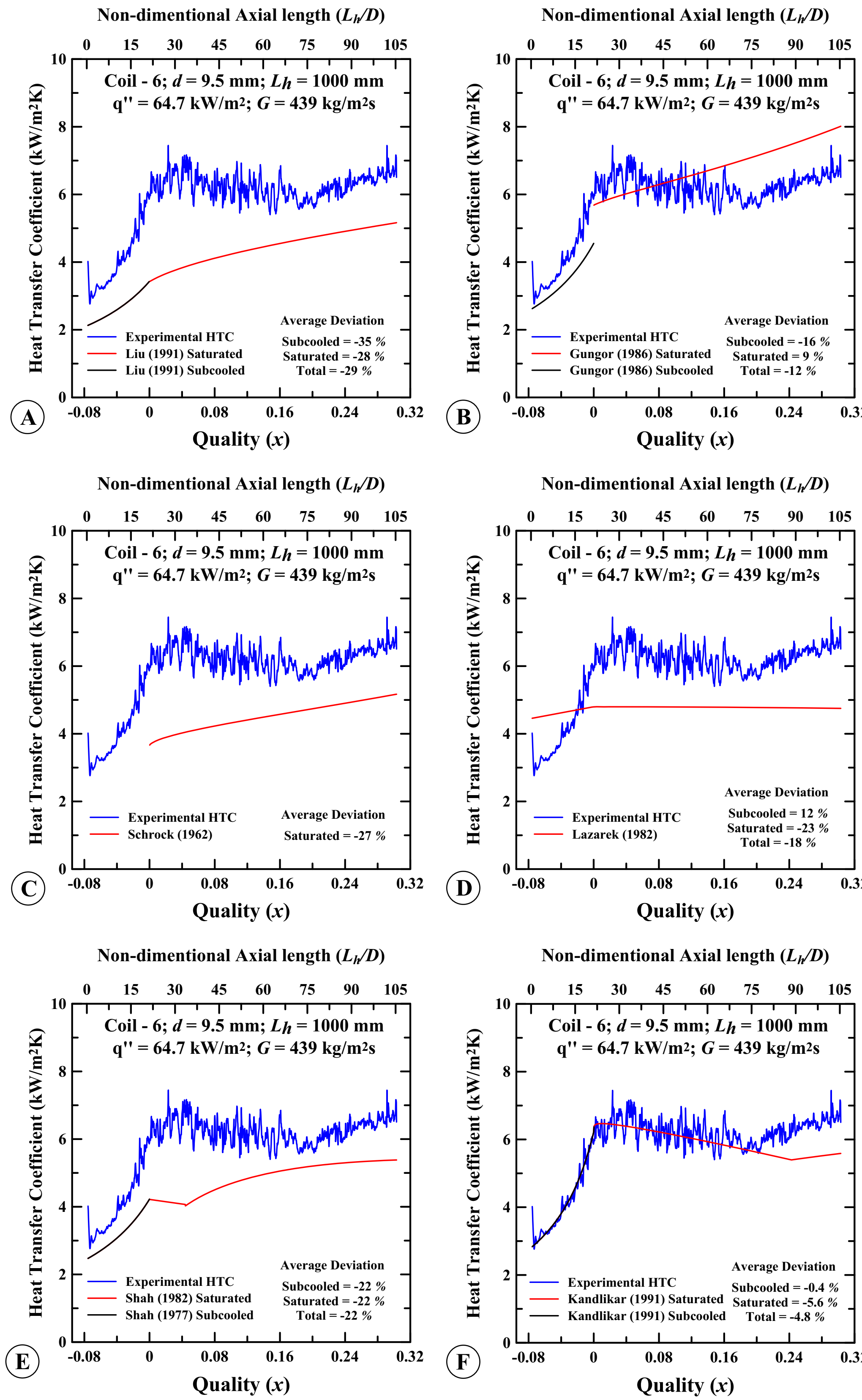

Fig. 6. Comparison of local heat transfer coefficient with available correlations.

correlation is solved for critical heat flux with iterative method. The correlation predicts the experimental data with good accuracy. Well referred correlation of Merilo [22] predicts the experimental results with reasonable accuracy. The statistical comparison between measured data and data predicted using correlations is given in Table 5. Analysis shows that the correlation developed

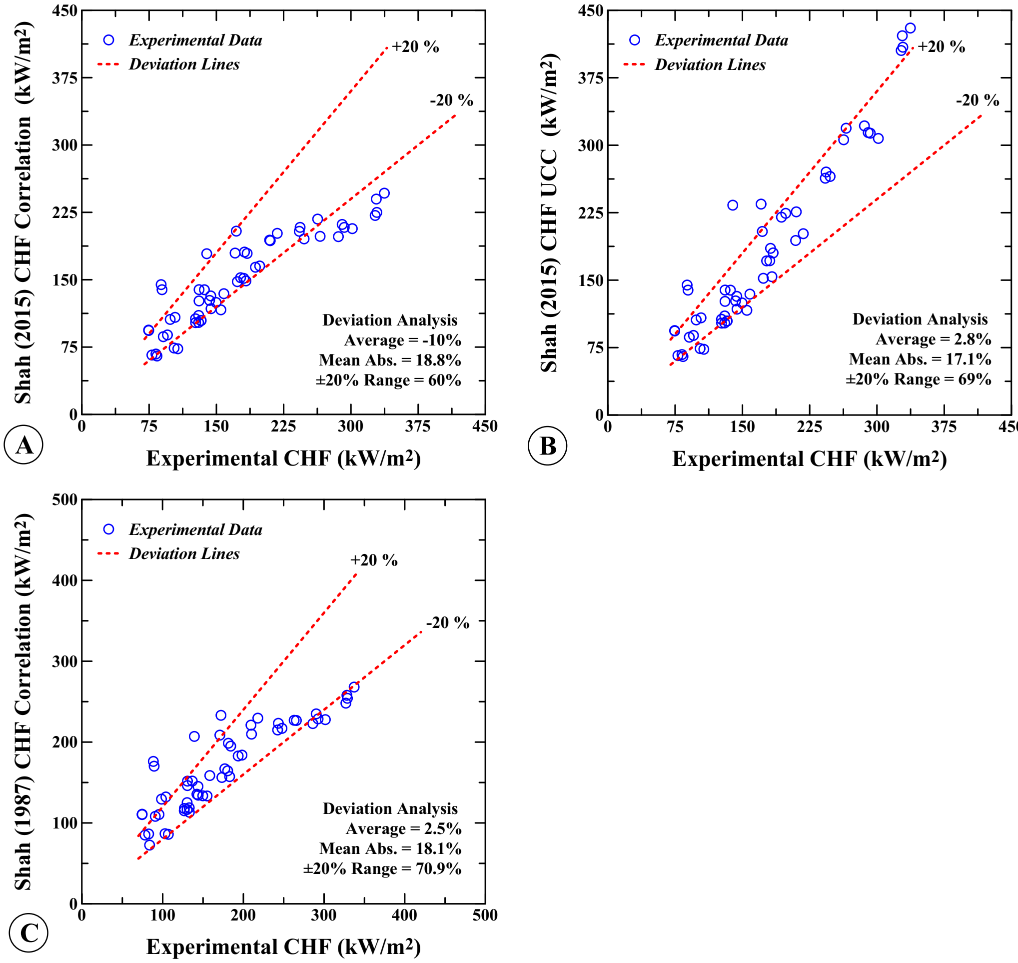

Fig. 7. Comparison of experimental critical heat flux data with Shah's correlations.

by Kim and Mudawar [18] match well with present experimental data followed by Merilo [22] and Shah [28] correlations.

4.6. Comparison of two-phase pressure drop with available correlations

Two phase pressure drop is measured along with the heat transfer coefficient in all the test sections given in Table 2. Parameter conditions of data collected for pressure drop and heat transfer coefficient are same. Total 171 data are collected for heat transfer and pressure drop. Out of these, the data of pure subcooled flow boiling are removed for two-phase pressure drop study. Apart from these data, few data are collected only for saturated flow boiling pressure drop analysis. Total 158 data are presented here for saturated two-phase pressure drop analysis. The data are compared with available correlations.

Fig. 9 shows the comparison of experimental two-phase pressure drop with the available correlations. The maximum and minimum deviation of data points which fall within are shown in the figure. The results show that the correlations underpredict the experimental pressure drop data. Chisholm [7] correlation overpredicts the data while most of the correlations underpredict the experimental data. Gronnerud [11] correlation is matching reasonably well with experimental data. The statistical comparison between measured data and data predicted using correlations is given in Table 6.

4.7. Suggested correlation for two-phase pressure drop

The present experimental data are compared with Chisholm [6] correlation as shown in Fig. 10(A). The correlation works well compared to other correlations. The Chisholm [6] correlation is under predicting the data. Most of the experimental data are deviating from 10% to −30%. Hardik and Prabhu [14] developed the correlation for two-phase pressure drop for water flow boiling as given in Eqs. (14)–(18). The correlation is developed based on the experimental data for tube diameter varying from 5.5 mm to 12 mm. The correlation is developed by adding the effect of tube diameter in Chisholm [6] correlation. The correlation is developed for operating parameter range; mass flux 100–1200 kg/m^2 s, heat flux 400–1800 kW/m^2, quality 0–0.65 and density ratio 550–1600. The same correlation is compared with present experimental data. The operating parameter range of present data is mass flux 262–1650 kg/m^2 s, heat flux 25–337 kW/m^2, quality 0–0.86 and density ratio 75–175. The comparison is show in Fig. 10(B). Hardik and Prabhu [14] correlation works well for present data. The correlation is predicting the present data within ±20% deviation. Hence, the correlation may be used for general purpose to predict the two-phase pressure drop with operating parameter range mass flux 100–1650 kg/m^2 s, heat flux 25–1800 kW/m^2, quality 0–0.86 and density ratio 75–1600.

$$\Delta P_{fric} = \varnothing^2_{Ltt} \Delta P_l \tag{14}$$

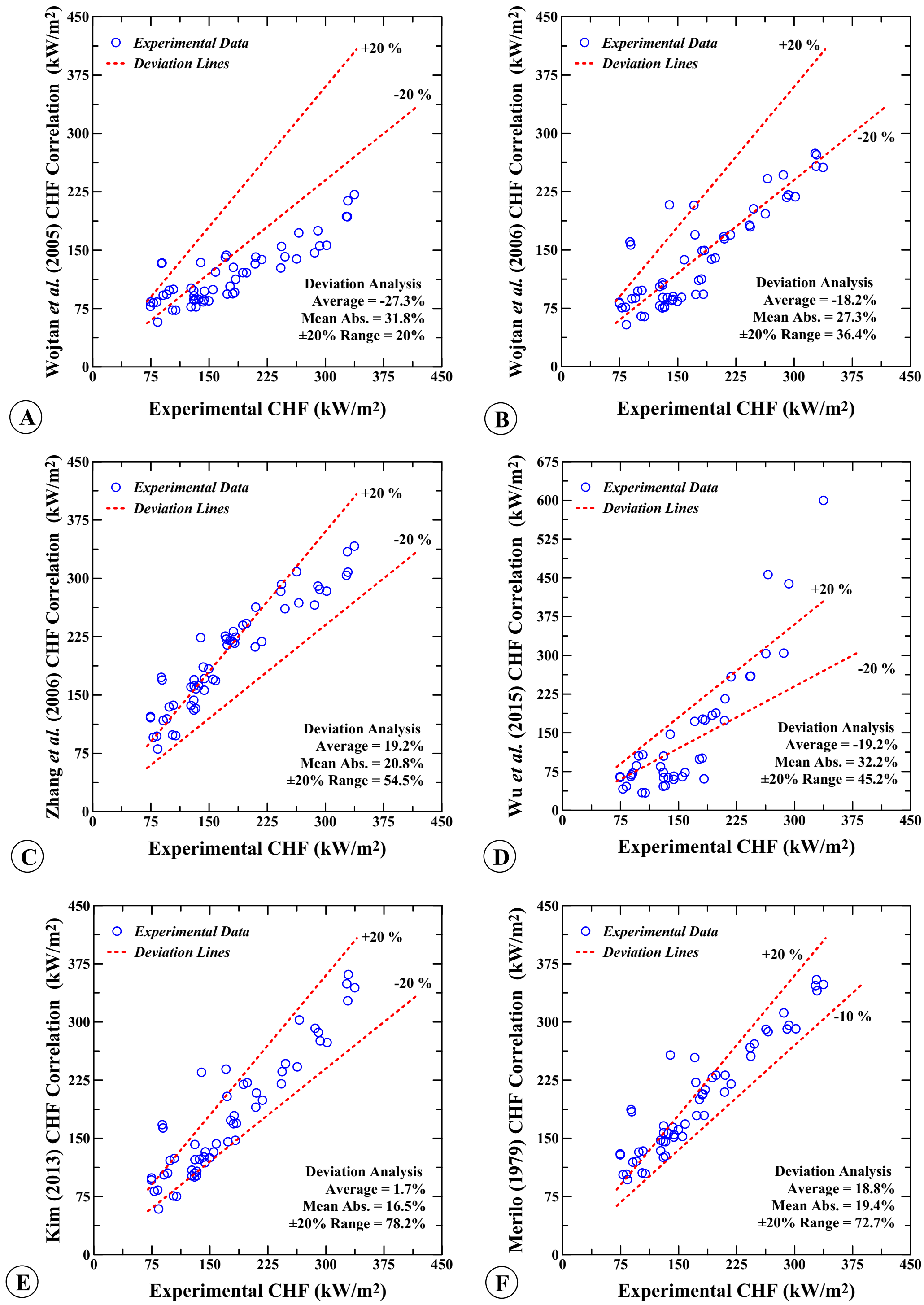

Fig. 8. Comparison of experimental critical heat flux data with available correlations.

Table 5
Statistical analysis of experimental CHF data with available correlations.

Authors (Year)	Mean average	Absolute mean	±10%	±20%	±30%
Kim and Mudawar [18]	1.66	16.5	45.5	78.2	90.9
Merilo [22]	18.8	19.4	49.1	72.7	85.5
Shah [28] UCC	2.77	17.1	36.4	69.1	90.9
Shah [27]	2.47	18.1	36.4	70.9	87.3
Shah [28]	−10.0	18.8	25.5	60.0	87.3
Zhang et al. [39]	19.2	20.8	40.0	54.5	83.6
Wojtan et al. [33]	−18.2	27.3	16.4	36.4	54.5
Wu et al. [35]	−19.2	32.2	31.0	45.2	52.3
Wojtan et al. [34]	−27.3	31.8	14.5	20.0	29.1

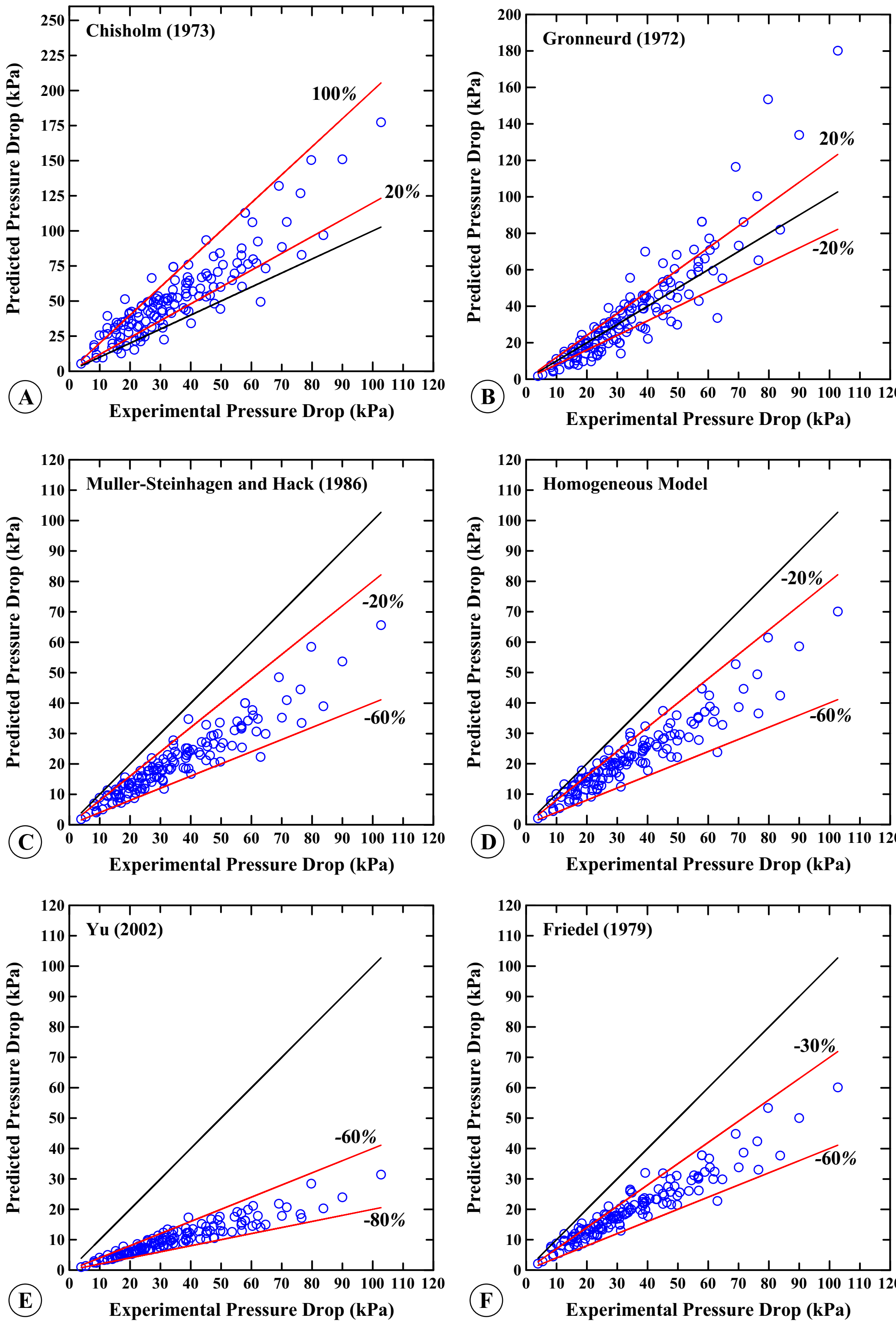

Fig. 9. Comparison of experimental two-phase pressure drop with available correlations.

$$\varnothing_{ltt}^2 = 1 + \frac{C}{X_{tt}} + \frac{1}{X_{tt}^2} \tag{15}$$

$$C = 18 \times e^{(0.14\times d)} \tag{16}$$

$$\Delta P_l = \frac{2 f_l L G^2 (1-x)^2}{d \rho_l} \tag{17}$$

$$f_l = \frac{0.079}{Re^{0.25}}; Re = \frac{Gd}{\mu_l} \tag{18}$$

5. Conclusions

Experiments are performed to study heat transfer coefficient distribution, pressure drop and critical heat flux in horizontal

Table 6
Deviation of correlations from the experimental pressure drop.

Sr. No.	Author	Absolute mean	Average	Maximum range
1	Hardik and Prabhu [14]	14	4	−30 to 52
2	Chisholm [6]	17	−10	−44 to 46
3	Gronnerud [11]	23	−2	−57 to 93
4	Homogeneous Model	34	−34	−62 to 6
5	Muller-Steinhagen and Heck [25]	38	−38	−65 to −9
6	Friedel [10]	39	−39	−64 to −5
7	Zhang et al. [40]	41	−40	−69 to 5
8	Mishima and Hibiki [23]	41	−41	−70 to 5
9	Sun and Mishima [31]	43	−41	−72 to 6
10	Zhang and Webb [38]	43	28	−37 to 166
11	Borishansky et al. [3]	49	−31	−78 to 45
12	Tran et al. [32]	52	−26	−86 to 39
13	Chisholm [7]	56	54	−28 to 214
14	Chien and Ibele [5]	57	−57	−93 to −14
15	Xu and Fang [36]	58	32	−60 to 94
16	Beattie and Whalley [2]	66	−66	−90 to −33
17	Yu et al. [37]	70	−70.1	−78 to −56

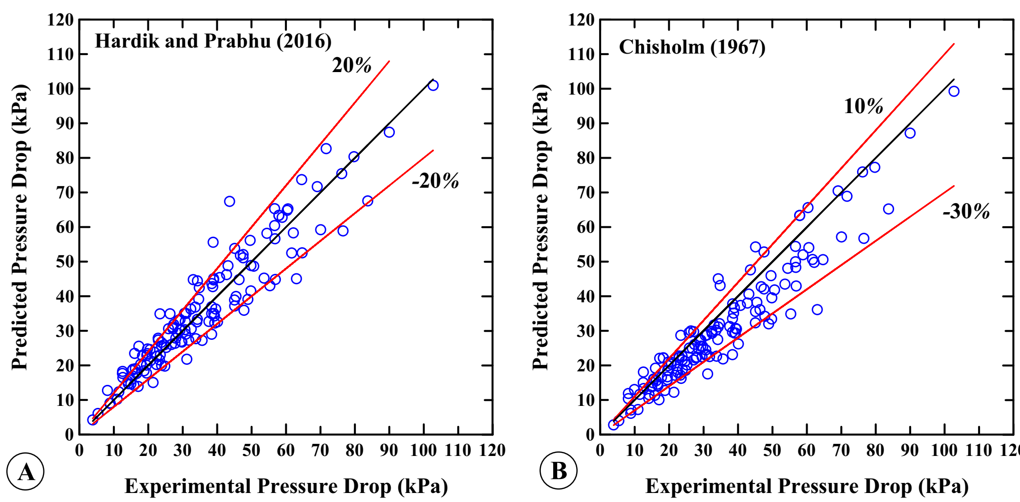

Fig. 10. Comparison of experimental two-phase pressure drop with suggested correlations.

straight tubes. Experiments are conducted for seven test sections with four tube diameters 5.5 mm, 7.5 mm, 9.5 mm and 12.0 mm. Local heat transfer coefficient is compared with the available correlations to check the capability of a given correlation to represent the boiling curve. Pressure drop, CHF and overall averaged heat transfer coefficient are compared with the available correlations. Following are the conclusions that may be drawn from the present study.

- Flow boiling curve is affected with the liquid to vapor density ratio. With increase in liquid to vapour density ratio, the length of nucleate boiling region increases and slope of the heat transfer coefficient in the convective region decreases.
- Overall averaged heat transfer coefficient data match well with Gungor and Winterton [12] correlation. However, Gungor and Winterton [12] correlation not represent actual boiling process. The correlation shows continuous increase in heat transfer coefficient in nucleate and in convective boiling region. While actually heat transfer coefficient decreases in the nucleate boiling region and increases in the convective boiling region. Moreover, the correlations for subcooled and saturated boiling heat transfer failed to match at zero quality.
- Kandlikar [17] correlation matches well with the overall averaged heat transfer coefficient data. It also represents the actual boiling process. The correlation shows decrement in nucleate and increment in convective boiling region same as actual heat transfer coefficient.
- Hardik and Prabhu [14] correlation matches well with present pressure drop data. The correlation originally derived with water data. Applicability of the same correlation for present data shows the general use of the correlation.
- CHF data of present study match well with Kim and Mudawar [18] and Merilo [22] correlations.

6. Contributions of the present work

The work presents a technique to measure the local wall temperature through infrared thermal imaging technique. The technique helps to calculate local heat transfer coefficient and to represent the actual boiling process in different regions of flow boiling. Local distribution of wall temperature and heat transfer coefficient along the axial length aids in identifying the different regions of flow boiling. The present work identifies the correlation which works well to predict subcooled and saturated heat transfer coefficient in a circular tube. The local and overall averaged heat transfer coefficient comparison with correlations helps researchers to differentiate the applications of different correlations. A correlation for two-phase pressure drop and CHF is suggested in order to accurately calculate the boiling pressure drop and CHF. The published CHF data helps to generate the data bank for CHF in horizontal straight tube. It also helps to develop a general correlation with better accuracy of prediction. Local heat transfer coefficient distribution presented in this study would serve as benchmark data for the validation of numerical results of flow boiling.

Conflict of interest

The authors declare that there is no conlict of interest.

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Appendix A

Experimental critical heat flux data.

Sr. no.	Tube dia. mm	Heated length mm	System pressure bar	Inlet temperature °C	Mass flux kg/m^2 s	Exit quality	CHF kW/m^2
1	5.5	500	1.99	33.5	400.5	0.624	126.8
2	5.5	500	1.83	32.5	431.5	0.577	130.3
3	5.5	500	2.17	33.2	835.2	0.375	184.3
4	5.5	500	2.46	33.0	854.3	0.346	181.4
5	5.5	500	2.71	33.7	1054.9	0.299	210.1
6	5.5	500	2.48	33.4	1255.8	0.289	242.4
7	5.5	500	2.95	33.9	1281.7	0.260	243.4
8	5.5	500	2.65	34.1	1479.3	0.245	262.9
9	5.5	1000	2.02	32.8	416.4	0.810	83.8
10	5.5	1000	2.59	33.7	880.0	0.548	133.1
11	5.5	1000	2.59	33.7	898.0	0.501	126.7
12	5.5	1000	2.82	33.4	1055.9	0.485	149.6
13	5.5	1000	2.78	33.4	1078.7	0.443	142.4
14	5.5	1000	3.12	34.5	1312.1	0.427	173.4
15	5.5	1000	3.19	34.6	1327.1	0.450	183.0
16	7.5	500	1.83	31.7	448.1	0.546	158.4
17	7.5	500	2.05	32.3	665.9	0.462	209.4
18	7.5	500	2.04	29.3	682.7	0.389	217.9
19	7.5	500	2.08	28.2	687.5	0.282	172.2
20	7.5	500	2.17	30.4	979.0	0.328	247.8
21	7.5	500	2.39	31.0	1151.5	0.330	301.6
22	7.5	500	2.55	29.9	1162.0	0.251	290.3
23	7.5	500	2.37	31.0	1175.6	0.316	292.6
24	7.5	500	2.64	31.2	1619.8	0.160	328.3
25	7.5	500	2.92	31.2	1643.8	0.152	337.3
26	7.5	1000	2.04	29.9	381.8	0.849	107.0
27	7.5	1000	2.03	29.9	386.0	0.820	102.9
28	7.5	1000	2.32	29.5	531.8	0.722	130.2
29	7.5	1000	2.31	29.5	543.1	0.704	132.9
30	7.5	1000	2.20	29.0	684.1	0.492	143.8
31	7.5	1000	2.22	29.0	719.7	0.488	136.7
32	7.5	1000	2.19	29.0	720.3	0.461	130.7
33	7.5	1000	2.60	29.9	753.1	0.549	155.0
34	7.5	1000	2.54	29.8	763.9	0.492	144.0
35	7.5	1000	2.66	30.0	1152.9	0.337	177.0
36	7.5	1000	2.44	30.0	1163.2	0.350	180.5
37	9.5	500	1.62	28.8	406.2	0.197	89.4
38	9.5	500	1.65	29.6	427.6	0.187	88.4
39	9.5	500	1.77	29.2	712.3	0.158	139.3
40	9.5	500	1.85	29.4	714.5	0.202	170.9
41	9.5	500	2.35	32.1	1005.2	0.212	265.7
42	9.5	500	1.98	29.3	1007.5	0.247	286.0
43	9.5	500	2.36	29.1	1282.8	0.172	327.0
44	9.5	500	2.55	29.1	1289.2	0.162	328.9
45	9.5	1000	1.82	29.1	424.3	0.355	74.6
46	9.5	1000	1.71	30.3	432.5	0.364	74.5
47	9.5	1000	2.10	31.7	707.6	0.370	130.6

Appendix A (*continued*)

Sr. no.	Tube dia. mm	Heated length mm	System pressure bar	Inlet temperature °C	Mass flux kg/m^2 s	Exit quality	CHF kW/m^2
48	9.5	1000	2.70	31.4	1272.8	0.231	193.5
49	9.5	1000	2.69	31.5	1301.1	0.233	198.3
50	12	1000	1.67	29.5	262.6	0.518	78.1
51	12	1000	1.69	29.6	267.3	0.540	82.6
52	12	1000	1.82	31.2	350.4	0.438	90.8
53	12	1000	1.85	31.3	358.8	0.450	95.4
54	12	1000	1.96	33.1	435.2	0.371	98.5
55	12	1000	1.99	33.1	445.2	0.383	103.9

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